

Swaroop Bhowmik

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Education

B.Tech, Computer Science Engineering (AI & ML), Vellore Institute of Technology Bhopal 2023 – 2027
CGPA: 8.62/10

Technical Skills

- **AI/ML Frameworks:** LangGraph, LangChain, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, NumPy, Pandas
- **Deployment & Infra:** FastAPI, ChromaDB, Ollama, Docker, REST APIs, SSE, Git
- **Concepts:** RAG Pipelines, Multi-Agent Systems, Prompt Engineering, Model Optimization, Semantic Segmentation, Image Classification
- **Languages:** Python, SQL

Projects

Tarka AI – Multi-Agent Research System [[Live Demo](#)] Jan 2026 – May 2026

LangGraph, LangChain, LLMs, Tavily API, FastAPI, SSE, ChromaDB, Pydantic, Docker

- Designed a LangGraph StateGraph orchestrating five specialized agents with conditional routing and an iterative Critic→Searcher loop capped at 3 iterations to close knowledge gaps without unbounded search cost.
- Built a custom reranker (relevance score + query-term overlap, with domain-diversity bonuses) and a ChromaDB semantic cache (0.15/0.55 cosine thresholds) that cut redundant LLM calls on repeated queries to near-zero cost.
- Implemented structured, confidence-scored claim extraction with source citations, and exposed it via SSE-streaming FastAPI endpoints for real-time UI feedback; deployed via Docker Compose with health checks.

AI-Powered Medical Chatbot (RAG Architecture) [[GitHub](#)]

Dec 2025 – Feb 2026

FastAPI, LangGraph, RAG, LLMs, Ollama, ChromaDB, REST APIs

- Built a medical QA chatbot using RAG over large PDF datasets, with an agentic LangGraph workflow for retrieval, relevance grading, and hallucination detection.
- Deployed fully locally with Ollama, ChromaDB, and FastAPI for cost-free inference with no external API dependency.

Satellite & Indoor/Outdoor Image Dehazer [[GitHub](#)]

Oct 2024 – Apr 2025

Python, TensorFlow, Keras, NumPy, Pandas

- Designed a CNN-based cGAN with a novel Cross-Scale Feature Fusion Block, enabling single-sensor RGB satellite dehazing without SAR data.
- Achieved SOTA results (PSNR 26.1, SSIM 0.946) on Haze1k, outperforming GridDehazeNet and SAR-Opt-cGAN with validated ablation studies. Paper Under Review

CED based Dual Environment Image Dehazer [[GitHub](#)]

Oct 2023 – Jan 2024

Python, TensorFlow, Keras, NumPy, Pandas, cGAN

- Built an Enhanced U-Net (dilated convolutions, Leaky ReLU) for indoor/outdoor dehazing, achieving a 2 dB PSNR gain over baseline U-Net and a 4% SSIM improvement over 9 SOTA methods (DCP, AOD-Net, DehazeNet) on Dense-Haze NTIRE19. [[Published Paper](#)]

Research Publications

- Dual-Environment Dehazing Using Enhanced U-Net for Indoor and Outdoor Scenarios, IEEE SCEECs 2025. [DOI](#)
- Hybrid Statistical-Neural Framework for Context-Aware Word Sense Disambiguation Using Multi-Head Attention and TF-IDF Fusion. [DOI](#)
- Enhanced Satellite Image Dehazing using a CNN-Based cGAN with Cross-Scale Feature Fusion Block (CSFFB) (Under Review)

Achievements

- Raman Research Award, VIT Bhopal, 2025 – for IEEE-published research. [Google Scholar](#)
- Certificate of Presentation, NEXGCI-2025, SCSET, Bennett University.
- Granted Design Registration (No. 449246-001) by the Patent Office, Government of India for a Multipurpose Vegetable Chopping Board under the Designs Act, 2000.

Certifications

- Python for Data Science, AI & Development (IBM)
- Cloud Computing (IBM)
- Generative AI (Oracle)
- Applied Machine Learning in Python (Uni of Michigan)